1) Benchmark Construction

Source QA Datasets

- H HotpotQA
- M MuSiQue
- S SQuAD

Base QA Sample

Q: Where was the founder of Tesla born?

A: Pretoria, South Africa.

Evidence (supporting facts):
- Elon Musk is the founder of Tesla.
- Elon Musk was born in Pretoria, South Africa.

Variant Generation

- Position Manipulation front / middle / end
- Conflict Manipulation wrong-before-correct / correct-before-wrong
- Distractor Injection distractor-dominant

6 Variants per Base Sample

1 2 3 4 5 6

1 2 3 Position Variants

4 5 Conflict Variants

6 Distractor-dominant

2) Example: 6 Evidence-Layout Variants

1 Correct-Front (CF)

[Correct] Elon Musk was born in Pretoria, South Africa.	**Distractor** Tesla was founded in 2003.	**Distractor** Tesla is an electric car company.

2 Correct-Middle (CM)

Distractor Tesla stock price rose in 2020.	**Distractor** Tesla Model S was launched in 2012.	**[Correct]** Elon Musk was born in Pretoria, South Africa.	**Distractor** Tesla has factories in USA and China.

3 Correct-End (CE)

Distractor Tesla builds electric cars.	**Distractor** Elon Musk is the CEO of SpaceX.	**Distractor** Tesla Energy produces solar products.	**[Correct]** Elon Musk was born in Pretoria, South Africa.

4 Conflict-Before-Correct (CBC)

[Wrong] Elon Musk was born in Canada.	**Distractor** Tesla was founded in 2003.	**Distractor** He studied physics and economics.	**[Correct]** Elon Musk was born in Pretoria, South Africa.

5 Correct-Before-Conflict (CFC)

[Correct] Elon Musk was born in Pretoria, South Africa.	**Distractor** Tesla builds electric cars.	**Distractor** He is known for SpaceX and Tesla.	**[Wrong]** Elon Musk was born in Canada.

6 Distractor-Dominant (DD)

Distractor ...	**Distractor** ...	**Distractor** ...	**Distractor** ...	**[Correct]** Elon Musk was born in Pretoria, South Africa.

Correct Evidence · Wrong Evidence · Distractor Evidence

3) Multi-LLM Evaluation

LLM Pool

GPT · Claude · Gemini · DeepSeek

Bias Phenomena under Test

- **Position Bias** Does evidence position (front / middle / end) affect model performance?
- **Conflict Bias** Does conflicting evidence mislead the model depending on order?
- How do models respond to a distractor-heavy layout without planted conflicting evidence?

Bias-Sensitive Metrics

Acc	Answer Accuracy
PSR	Position Sensitivity Rate
CAR	Conflict Arbitration Rate
EAR	Error Adoption Rate
CEU	Correct Evidence Usage

Quantify & Compare RAG Bias

Cross-model comparison, statistical testing (Bootstrap), and in-depth analysis.